Exercise 04: *cocotb* Test

Objective: *Introduction to the cocotb tests.*

Task: *Development of a cocotb test using the Saturation Filter.*

In this exercise you need to develop a simple *cocotb* testbench for a saturation filter (SAT Filter). First make yourself familiar with the DUT. Do not look at the RTL, but rather read the provided documentation in `lecture03/exercises/E04_sat_cocotb_test/exercise_general_info.pdf`.

Part 1: Simple Test

`lecture03/exercises/E04_sat_cocotb_test/tb` contains a `Makefile` which already correctly imports the SAT filter RTL from `lecture03/exercises/E04_sat_cocotb_test/rtl`. You only need to update `COCOTB_TEST_MODULES` with you test module name.

Create a new *cocotb* test with a single testcase, which:

- Starts a clock on `clk`.
- Resets the DUT using `rst`.
- Sends two transaction through the DUT.
  - A transaction with a value below the `THRESHOLD`.
  - A transaction with a value above the `THRESHOLD`.
  - Be sure to adhere to the *sSDT* Protocol.
- Check the received outputs using the `assert` statement.

Implement your test in the file: `sat_default_test.py` and add it to the `Makefile`s `COCOTB_TEST_MODULES` variable.

Run the test using the `Makefile` and inspect the waveforms.

Part 2: Multiple transactions

Extend the testcase to drive multiple transactions.

- Generate 10 sets of random input data using pythons `random` module;
- Loop over the generated input data. In each iteration:
  - Send the transaction though the DUT.
  - Check the received output using the `assert` statement.

Observe the test results and the waveforms.